

Bob AutoHeal

Automated Testing & Validation Hub

Built for the IBM Bob 2.0 Hackathon

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The Challenge

The Manual Debugging Bottleneck

Software teams waste **30–45 minutes** per bug on a repetitive, manual loop:

**Write Code →
Push → CI Fails
→ Read Logs →
Guess Root
Cause → Fix →
Repeat**

- **Slow:** Requires human context switching
- **Error-Prone:** Developers often patch symptoms, not root causes
- **Unscalable:** As codebases grow, bugs outpace the team

The Solution

Bob AutoHeal Workflow

Using IBM Bob 2.0 to fully automate the **Detect → Diagnose → Fix → Verify** loop.

1. **Validation:** Run test suite (pytest) & capture coverage
2. **Research:** Bob analyzes stack traces & source code
3. **Healing:** Bob dispatches parallel Coder Subagents to fix bugs

IBM Bob 2.0 Utilization

Leveraging the Full Agentic Stack



Agent Mode: Orchestrates the entire validation and healing loop autonomously based on custom Skill instructions.



Subagents: Spawns a dedicated Research Subagent for analysis, and Coder Subagents for precise fixes.



Parallel Tasks: All bug fixes are dispatched simultaneously to Coder subagents, cutting resolution time by ~3x.



Document Understanding: Reads XML coverage reports, Python source code, and pytest stack traces to cross-reference errors.

Business Impact

Before vs. After AutoHeal

Metric	Manual Debugging	Bob AutoHeal
Test Success Rate	62.5% (Failing)	100% (Passing)
Code Coverage	73%	100%
Time to Diagnose	~20 mins	~30 seconds
Time to Fix 3 Bugs	~25 mins	~90 seconds (Parallel)
Total Cycle Time	~45 minutes	~2 minutes